

Haoyu (Novak) Chen

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EDUCATION

Columbia University, New York City, NY

Master of Arts in Statistics, GPA: 3.91/4

September 2023 – December 2024

Advisor: Dr. Liam Paninski

- **Coursework:** Advanced Machine Learning, Multi-level Bayesian Regression, Python for Deep Learning, Statistical Analysis of Neural Data

Clark University, Worcester, MA

Bachelor of Arts in Mathematics and Geographic Information Systems, GPA: 3.78/4

August 2017 – June 2021

Advisor: Michael Satz

- **Honors:** Magna Cum Laude; Recognized for contribution to George Perkins Marsh Institute at Clark University
- **Coursework:** Probability and Statistics | Linear Models | Stochastic Processes | Differential Equations and Analysis |
- **Project:** Conducted inter-department joint honors thesis using random walk theory study African elephant movement using satellite geographic locations.

RESEARCH EXPERIENCE

Research Assistant II

Dr. Attila Losonczy, The University of Texas Southwestern Medical Center

September 2025 – Present

Dallas, Texas

- Co-developed neural decoding and topological analyses for *NeuralFieldManifold* (accepted, NeurIPS 2026), investigating how delay-embedded neural signal geometry relates to sleep states and motor behavior; validated analyses through benchmarking and feature ablations.
- Led a cross-modal registration pipeline aligning two-photon in vivo and confocal volumetric microscopic images of the same dendritic arbors, consisting of denoising, 3D segmentation and 3D registration, allowing for high-throughput alignment across imaging modalities.
- Developing a modality-agnostic event-detection pipeline for calcium, voltage, and glutamate activity traces using self-supervised time-contrastive learning, temporal neural networks, and Bayesian state-space representations.
- Maintained a custom behavioral and virtual reality infrastructure integrating microscopy, visual stimulation and reward systems for closed-loop rodent learning experiments.
- Built analysis pipelines synchronizing two-photon imaging, pose estimates, behavioral event logs, and TTL signals for head-fixed and freely moving mouse experiments.
- Supported mouse behavioral training, handling, and surgical preparation and analysis for ongoing neuroscience studies.

Research Assistant

Dr. Rene Hen, Department of Psychiatry, Columbia University

January 2025 – August 2025

New York, NY

- Led in building a behavioral phenotyping pipeline for the stress-enhanced fear learning (SEFL) paradigm, combining classical analyses of freezing behavior with pose tracking (DeepLabCut) and unsupervised syllable segmentation (Keypoint-MoSeq) to identify behavioral correlates of PTSD susceptibility across age, sex, and fear-learning conditions; cross-validated by using extracted behavior motifs as features in classifiers to predict fear-learning phenotype. *Manuscript in preparation.*
- Applied DeepLabCut and Keypoint-MoSeq pipeline to ventral CA1 projection imaging data to identify generalizable "anxiety-associated" behavioral hidden states across multiple anxiety assays (open field test, elevated plus maze) and examine motifs whose frequency anti-correlates with exploration of exposed zones. *Contributing to manuscript in preparation.*
- Supported an R21 grant submission with EEG- and EMG-based sleep-stage classification analysis, power analysis, and figure preparation.
- Developed modular, class-based analysis pipelines on GitHub for version control, maintaining branches, tracking issues, and creating/reviewing pull requests to ensure code quality and collaboration.

Research Assistant

Dr. Yuan Zhang, Columbia Aging Center, Mailman School of Public Health, Columbia University.

June 2024 – December 2024

New York, NY

- Conducted statistical analysis of large-scale aging and cognitive datasets, examining the influence of social and life-course factors on health outcomes

- Collaborated with the Harmonized Cognitive Assessment Protocol (HCAP) network through the University of Michigan Health and Retirement Study, an NIA-funded longitudinal cohort.

Research Assistant

September 2023 – December 2024

Dr. George Gushue, Social Cognition and Career Lab at Teachers College, Columbia University New York, NY

- Performed power and path analysis for projects examining sociocultural and cognitive predictors of academic persistence
- Assisted with study design and IRB preparation, contributing to a project presented at the national Latinx Psychological Association Conference

Project Manager

December 2022 – August 2023

Developmental Psychology, Boston College

Boston, MA

- Led data collection and analysis for a multi-site developmental psychology project, including statistical modeling and hypothesis testing (ANOVA, regression, SEM) using R and SPSS
- Major contribution to multiple peer-reviewed publications and conference presentations resulting from this work

PUBLICATIONS & PRESENTATIONS

Fallah, K., **Chen, H. N.**, Singha, R., Kong, E., Turi, G., Losonczy, A., Zabeh, E. (2026). NeuralFieldManifold: Reconstruction of LFP Manifold with Lag Embedding. *NeurIPS 2026*. Accepted.

Leal Santos, S., Lim, S. C., Chang, W., Chung, H., **Chen, H.**, Jimenez, J., Lacefield, C. O., Turi, G., Fusi, S., & Hen, R. (2025, September 16–18). Population codes for anxiety in the ventral hippocampus. Poster presented at the Hope for Depression Research Foundation (HDRF) Annual Scientific Summit, Durham, North Carolina, USA

Khan S, Turi G, Chung H, **Chen N**, Hen R, Chang WL. Aging Effects on Serotonergic Modulation of the Dentate Gyrus during Fear Learning and Pattern Separation in Mice. Society of Biological Psychiatry Annual Meeting. Toronto, ON, April 24-26, 2025, #S101.

PUBLICATIONS & PRESENTATIONS

Chen, H. (2025, November). Attendee, Society of Neuroscience Annual Meeting (SfN).

Chen, H. (2025, July). Attendee, Neuroscience and Artificial Intelligence Seattle Meeting. Allen Institute and University of Washington, Seattle, WA.

TEACHING EXPERIENCE

Teaching Assistant - Linear Algebra

September 2020 – December 2020

Clark University Department of Mathematics

Worcester, MA

- Held weekly teaching sessions to provide students help with lectures and homework
- Evaluated weekly assignments and provided feedback to each student

SKILLS

Coding: R, Python, MATLAB, SPSS, ArcGIS products; Intermediate in SQL, Tableau

Software: Suite2p, SIMA, OASIS, napari, DeepLabCut, SLEAP, Lightning Pose, Keypoint-MoSeq

Languages: Fluent in Mandarin and English